ACN: 1227329

REPORT CLASSIFICATION

Classification Type	Full Form Analysis

DATE / TIME

Date of Occurrence	2014-12
Local Time Of Day	1801-2400

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Light	Night

AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737 Next Generation Undifferentiated
Operating Under FAR Part	Part 121
Flight Phase	Parked
Maintenance Status - Maintenance Deferred	N
Maintenance Status - Maintenance Type	Scheduled Maintenance
Maintenance Status - Maintenance Items Involved	Inspection
Maintenance Status - Maintenance Items Involved	Work Cards

COMPONENT 1

Aircraft Reference	X
Aircraft Component	Fuselage Bulkhead
Manufacturer	Boeing
Problem	Design
Problem	Malfunctioning

COMPONENT 2

Aircraft Reference	X
Aircraft Component	Fuselage Skin
Manufacturer	Boeing
Problem	Failed

PERSON 1

Location Of Person	Gate / Ramp / Line
Reporter Organization	Air Carrier
Function - Maintenance	Technician
Qualification - Maintenance	Airframe
Qualification - Maintenance	Powerplant
Experience - Maintenance Technician	30 yrs
ASRS Report Number	1227329
Callback	Completed

EVENTS

Anomaly	Aircraft Equipment Problem - Critical
Anomaly	Deviation / Discrepancy - Procedural - FAR
Anomaly	Deviation / Discrepancy - Procedural - Published Material / Policy
Detector - Person	Maintenance
Were Passengers Involved In Event	No
When Detected	Routine Inspection
Result - General	Maintenance Action

ASSESSMENTS

Contributing Factors / Situations	Aircraft
Contributing Factors / Situations	Company Policy
Contributing Factors / Situations	Human Factors
Primary Problem	Human Factors

NARRATIVE 1

While completing my assigned Maintenance Visit Check, paragraph 20, on [a B737 NG aircraft,] I found a one and one quarter inch (1-1/4 inch) crack in the [Main] wheel well on the forward pressure bulkhead, Fuselage Station XXX Right Body Line (RBL) 40, approximately three inches down from the ceiling. I took pictures of the area for Engineering and had Inspection verify the crack using High-Frequency Eddy Current (HFEC). The crack was visible while standing in the right wheel well on the ground and without the use of stands or ladders. The crack is in an exposed chemical milled pocket on the bulkhead. The crack started in the radius of the chemical milled [section] and was continuing to follow the radius of the pocket, which, [if] left unchecked, would have resulted in a six inch diameter pocket blow out with a rapid pressurization loss event.

During my time on the Line Routine Overnight (RON) shift I have found numerous cracks, and regardless of the check being performed have reported them, to have them repaired and the planes returned to a SAFE and AIRWORTHY STATUS. Keeping our planes in a safe and airworthy condition is and has always been my number one priority at our Company. There is no Structural Repair Manual (SRM) or Engineering relief for this un-airworthy condition to continue in service without immediate repair. Our Maintenance Procedures Manual (MPM) XX-XX-XX states, "Any abnormalities or defects are recorded (written up) on the documents for inclusion in the work package". MPM XX-XX-XX states, "Regardless of the level of inspection accomplished, the priority is to ensure aircraft are maintained in a condition for continued safe operation." Other AMT's at this and other stations are finding these defects at all levels of checks on overnight inspections.

I am concerned that fear of being brought to an investigation with possible disciplinary actions will only deter AMT's from finding and reporting UN-AIRWORTHY Conditions, resulting in UNSAFE aircraft being returned to service to carry passengers.

CALLBACK 1

Reporter stated that cracks in the pressure vessel are a No-Go item. Deferral is available from Boeing or Engineers. Temporary repairs applying reinforcing doublers are acceptable, requiring about two shifts of work that allow the cracks to be deferred until the next Heavy Check when more intense inspections are accomplished in that area and a permanent repair is installed. However, Line Mechanics are finding these cracks outside of the Scheduled Heavy Checks because of the noticeable track lines, known as smoke lines. These lines leave black marks from cabin pressure leaking through the crack, also indicating the pressure bulkhead crack is expanding and contracting.

Reporter stated the center fuel tank is a wet tank, located on the forward side of the same main wheel well forward bulkhead that has the cracks. But, the area that is cracking is above the top of the fuel tank in a pressurized pocket, an air barrier area that runs the entire width of the fuselage wheel well, left to right; an area that can be seen and is accessible under the cabin floorboards.

Reporter stated cracks usually start at the center of the chemical milled pocket skin of the main wheel well forward bulkhead, which is a thinner skin and migrates to the edge of the individual pockets where the skin is thicker. But, that is not the type of crack found. The visual crack found started at the edge, in the radius where the chemical milled skin and thicker skin transition. The crack was following the outer edge of the chemical milled skin and around the pocket circumference, opening the skin up like a can opener does.

Reporter stated this is strictly a cabin pressurization issue that could become a major depressurization incident in flight. The Center Fuel Tank has nothing to do with the crack issue. Newer B737 NG aircraft do not have the thinner chemical milled skin in the crack areas. The reporter suspects the addition of the Winglets has added stress loads that were not reinforced in the chemical mill skin areas. Mechanics are required by the FAA and FARs to report defects they find, but so far, they have not had any support when they do report these cracks during Line Maintenance work.

SYNOPSIS

A Line Aircraft Maintenance Technician (AMT) reports about finding a crack at the forward pressure bulkhead, three inches down from the main wheel well ceiling on B737 NG aircraft. Noted cracks are a NO-GO item. Concerns raised that cracks might lead to rapid depressurization event.